

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and developed **Python**-based microservices using **FastAPI** to serve AI model inference endpoints, handling approximately **400 requests per second** during peak loads.
- Architected and optimized **data pipelines** using **Python** and **SQL** for ingesting and processing large datasets for AI model training, reducing data preparation time by **30%**.
- Integrated **LangChain** framework with internal knowledge bases to build a **RAG-based QA system**, improving response accuracy by **22%** based on user feedback.
- Containerized AI applications and backend services using **Docker** and deployed them to a scalable **AWS EKS** cluster, enabling automated rollouts and rollbacks via **Jenkins** pipelines.
- Developed and maintained high-performance **REST APIs** using **Java Spring Boot** for integrating AI model outputs into existing enterprise systems.
- Collaborated with product and engineering teams to define requirements and integrate **AI/LLM solutions** into core business workflows, ensuring seamless adoption.
- Monitored production AI systems using **AWS CloudWatch** and **Splunk**, establishing key performance indicators and responding to critical alerts to maintain system health.

SKILLS

AI/ML – LLMs, LangChain, RAG, Model Deployment, MLOps, Python, scikit-learn

Backend Development – Python, FastAPI, Java, Spring Boot, REST APIs, Microservices

Data Engineering – SQL, PostgreSQL, MySQL, Data Pipelines, ETL, Data Warehousing

Cloud & DevOps – AWS (EC2, S3, Lambda, EKS, CloudWatch), Docker, Kubernetes, Jenkins, Git

Observability – Splunk, Dynatrace

Data Analytics – Power BI, Tableau, DAX, Power Query, R, pandas, NumPy

PROJECTS

AI-Powered Document Analysis Service

- Developed a **Python** application using **FastAPI** to process and analyze unstructured documents, integrating **LLM capabilities** for summarization and keyword extraction.
- Implemented a **RAG pattern** to enable the service to query an internal document repository, improving the relevance of generated summaries by an estimated **15%**.
- Containerized the service using **Docker** for consistent deployment across development and production environments.

Customer Support Ticket Classification System

- Engineered a **Python**-based pipeline to extract features from customer support tickets using **pandas** and **SQL**, preparing data for model training.
- Trained a **scikit-learn** classifier (Naive Bayes) on a dataset of **~5,000 tickets** to categorize incoming requests, achieving an accuracy of **~0.82** on a held-out test set.
- Built a REST API endpoint using **Flask** to expose the classification model for integration into an internal ticketing system.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)